Abu Dhabi City Campus Daily/Weekly Planner

Week #: 8 (19/10/2026 To 23/10/2026)

Subject: Pre-Calculus Math **Unit Title: Functions, Graphs, Transformations, and Inverses** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Comparing and Classifying Parent Functions	• I can compare and classify parent functions using their graphs, equations, tables, and characteristics.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 4.2 Ensure access to tools and assistive technologies.	• Khan Academy: Parent Functions • Khan Academy: Inverse Functions • Desmos Graphing Calculator • Desmos Activity Builder
Lesson 2 Formative Assessment	• **Formative:** Lesson 1.5 (Parent Functions ONLY) Graph & recognize graphs of parent functions: Shapes, Domain, Range, End-Behavior, Any Vertices, Any Axis of Symmetry, Any Asymptote.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). Choose an item. Choose an item.	• Khan Academy • Desmos
Lesson 3 Revision	• **Revision for Summative Exam.**	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). Choose an item. Choose an item.	• Khan Academy • Desmos
Lesson 4 Summative Exam	• **Summative Assessment:** - Lessons 1.1 to 1.5 (Transformations are NOT included) - All slides and worksheets are posted on TEAMS - Videos on solving the worksheets, Graphing the parent functions and analyzing their characteristics have been posted on TEAMS.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). Choose an item. Choose an item.	• Khan Academy • Desmos
Homework	-	Choose an item. Choose an item. Choose an item.	
Teacher's notes/ Reminders	• Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments. • Bring a charged device for graphing and technology-based investigations. • Review vocabulary including function, relation, continuity, limit, extrema, and rate of change. • Practice explaining solutions using appropriate mathematical notation and justification.		

Technology Integration	• **Desmos.com**, Microsoft Teams, Graphing Calculator, GeoGebra
Cross-curricular links	**Physics:** Motion graphs and interpreting changing quantities. **Art and Design:** Symmetry, reflections, and scaling transformations. **Business Studies:** Multi-step cost, revenue, and profit models. **Computer Science:** Encoding, decoding, and reversible processes.
External Assessment Integration	Students will engage with: • **SAT/ACT/AP Precalculus Preparation:** Students analyze function behavior, interpret graphs, identify continuity, determine rates of change, and solve higher-order reasoning problems involving multiple representations.
Wellbeing/ ATL Skills	• **Critical Thinking:** Analyzing and comparing function behavior. • **Communication Skills:** Justifying mathematical conclusions. • **Research Skills:** Using graphing technology to investigate patterns. • **Self-Management:** Monitoring progress and reflecting on strategies. • **Thinking Skills:** Connecting graphical, numerical, and algebraic representations.
Standards	• **CCSS.HSF-IF.B.4** Interpret key features of graphs and relate them to quantities. • **CCSS.HSF-IF.C.7** Graph functions and show key features graphically. • **CCSS.HSF-IF.C.7a** Graph linear, quadratic, cubic, absolute value, square root, and exponential functions. • **CCSS.HSF-LE.A.1** Distinguish between linear and exponential growth. • **CCSS.HSF-BF.A.1** Build functions to model relationships.
Inquiry/ TOK Question	• To what extent do different function families help us model and predict patterns in the real world?
Success Criteria (Measurable)	**Topic 1: Function Notation, Domain & Range** ▨ All students can Identify functions using the vertical line test and evaluate functions using function notation. ◍ Most students should determine the domain and range of functions from graphs and equations. ▲ Some students could analyze more complex functions, including restrictions on domain and range, and justify answers mathematically. **Topic 2: Key Features of Graphs** ▨ All students can identify x- and y-intercepts, maximum and minimum points, and basic end behavior. ◍ Most students should determine intervals of increase and decrease, extrema, and end behavior from graphs. ▲ Some students could analyze and interpret multiple graph features, using correct mathematical notation and justification. **Topic 3: Continuity, End Behavior & Limits** ▨ All students can identify continuous and discontinuous functions and describe basic end behavior. ◍ Most students should estimate limits from graphs and classify discontinuities as removable or infinite. ▲ Some students could analyze one-sided limits, determine whether limits exist, and explain continuity at specific points.

	Topic 4: Average Rate of Change All students can calculate the average rate of change of a function over a given interval using the formula. Most students should interpret average rate of change as the slope of a secant line and apply it to different functions Some students could analyze changing rates over multiple intervals and connect average rate of change to instantaneous rate of change **Topic 5: Parent Functions** All students can recognize and sketch the six basic parent functions: linear, quadratic, cubic, absolute value, square root, and exponential Most students should determine domain, range, intercepts, symmetry, and end behavior of parent functions Some students could compare parent functions, justify their key features, and predict how transformations affect their graphs.
National Identity Domain	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث
	Area: UAE Innovation and Future Readiness Area: UAE culture and heritage Area: UAE Economic Development **Area:** Responsible Citizenship and Innovation
	Implementation: • Students examine how mathematical models are used in UAE smart-city projects and data-driven planning initiatives. • Students investigate geometric patterns and symmetry in traditional Emirati architecture and Islamic design. • Students analyze simplified models related to tourism growth, transportation systems, and business productivity in the UAE. • Students explore how inverse processes are used in digital systems, cybersecurity, and technological innovation supporting the UAE's vision for a knowledge-based economy.